Algorithm Manual - analysis in Go

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第一章 数论

1.1 GCD

```
package main
import "fmt"

func gcd(a ,b int) int {
    for {
        if b == 0 {
            return a
        }
        c := a%b
        a = b
        b = c
    }
}

func main () {
    fmt.Println(gcd(15, 12))
}
```

1.2 快速幂

```
package main
import "fmt"

func qpow(x, n int) int {
    if n == 0 {
        return 1
    }
    if n < 0 {
        x = 1 / x
        n = -1 * n
    }

    result := 1
    for {
```

```go
        if n == 0 {
            break
        }
        if n&1 == 1 {
            result *= x
        }
        x *= x
        n >>= 1
    }

    return result
}

func main () {
    fmt.Println(qpow(2, 10))
}
```

1.3 素数线性筛

```go
package main
import "fmt"

func findPrimes(n int) []int {
    var results []int
    isPrimeMap := make(map[int]bool)
    isPrimeMap[0], isPrimeMap[1] = false, false
    for i := 2; i<=n; i++ {
        if _, ok := isPrimeMap[i]; !ok{
            results = append(results, i)
        }
        for j:=i; j<=n; j+=i {
            isPrimeMap[j] = false
        }
    }
    return results
}

func main () {
    fmt.Println(findPrimes(100))
}
```

1.4 概率期望

第二章 排序

2.1 冒泡排序

```go
package main
import "fmt"

func max(a, b int) int {
    if a > b {
        return a
    }
    return b
```

```go
}

func bubbleSort(a[]int) []int {
    n := len(a)
    for i:=n-1; i>=0; i-- {
        for j:=0; j<i; j++ {
            if a[i] < a[j] {
                a[i], a[j] = a[j], a[i]
            }
        }
    }
    return a
}

func main () {
    a := []int{1, 3, 2, 5, 3, 7, 4, 5}
    fmt.Println(bubbleSort(a))
}
```

2.2 快速排序

```go
package main

import "fmt"

func quickSort(a []int, l int , r int) {
    if l >= r {
        return
    }
    x := a[l]
    i, j := l, r
    for i < j {
        for i<j && a[j]>=x {
            j-=1
        }
        if i < j {
            a[i] = a[j]
            i+=1
        }

        for i<j && a[i]<x {
            i+=1
        }
        if i < j {
            a[j] = a[i]
            j-=1
        }
    }
    a[i] = x
    quickSort(a, l, i-1)
    quickSort(a, i+1, r)
}

func main() {
    a := []int{5, 4, 3, 2, 1}
    quickSort(a,0, len(a)-1)
    fmt.Println(a)
```

```
    return
}
```

2.3 归并排序

2.4 堆排序

2.5 无序数组中位数

2.6 求两个有序数组的中位数或者第k小元素

2.7 全排列

第三章 数据结构

3.1 链表

链表结构

```
type ListNode struct {
    Val int
    Next *ListNode
}
```

求单链表中结点的个数

```
fun GetListLen(head *ListNode) int {
    var result 0
    for {
        if head == nil {
            break
        }
        head = head.Next
        result += 1
    }
    return result
}
```

将单链表反转

```
func reverseList(head *ListNode) *ListNode {
    if head == nil || head.Next == nil {
        return head
    }

    next := head.Next
    head.Next = nil

    for {
        if next.Next == nil {
            break
        }
```

```
        nextNext := next.Next

        next.Next = head
        head = next
        next = nextNext
    }
    next.Next = head

    return next
}
```

查找单链表中的倒数第K个结点（k > 0）

```
func getKthFromEnd(head *ListNode, k int) *ListNode {
    nodek := head
    for {
        if k == 0 {
            break
        }
        head = head.Next
        k -= 1
    }
    for {
        if head == nil {
            break
        }
        head = head.Next
        nodek = nodek.Next
    }
    return nodek
}
```

查找单链表的中间结点

```
func getMiddleNode(head *ListNode, k int) *ListNode {
    if head == nil || head.Next == nil {
        return head
    }

    faster := head
    for {
        if faster == nil || faster.Next == nil {
            break
        }
        head = head.Next
        faster = faster.Next
        if faster !=nil && faster.Next != nil {
            faster = faster.Next
        }
    }
    return head
}
```

从尾到头打印单链表

```
func reversePrint(head *ListNode) []int {
    var arr, result []int
    if head == nil {
        return result
    }

    for {
        now := head.Val
        arr = append(arr, now)
        head = head.Next
        if head == nil {
            break
        }
    }

    for i:=len(arr)-1; i>=0; i-- {
        result = append(result, arr[i])
    }
    return result
}
```

合并两个排序的链表

```
func mergeTwoLists(l1 *ListNode, l2 *ListNode) *ListNode {
    if l1 == nil {
        return l2
    }
    if l2 == nil {
        return l1
    }
    var result *ListNode
    if l1.Val < l2.Val {
        result = l1
        result.Next = mergeTwoLists(l1.Next, l2)
    } else {
        result = l2
        result.Next = mergeTwoLists(l1, l2.Next)
    }
    return result
}
```

判断一个单链表中是否有环

```
func hasCycle(head *ListNode) bool {
    if head == nil {
        return false
    }

    faster := head
    for {
        if faster == nil || faster.Next == nil {
            return false
        }
        faster = faster.Next.Next
        if head.Next == nil {
            return false
```

```
        }
        head = head.Next
        if faster == head {
            return true
        }
    }
    return false
}
```

两个单链表相交的第一个节点

```
func getIntersectionNode(headA, headB *ListNode) *ListNode {
    if headA == nil || headB == nil {
        return nil
    }

    a, b := headA, headB
    len_a, len_b := 1, 1
    for {
        if a.Next == nil {
            break
        }
        len_a += 1
        a = a.Next
    }
    for {
        if b.Next == nil {
            break
        }
        len_b += 1
        b = b.Next
    }
    if a != b {
        return nil
    }

    if len_a > len_b {
        count := len_a - len_b
        for count > 0 {
            count -= 1
            headA = headA.Next
        }
    } else if len_a < len_b {
         count := len_b - len_a
        for count > 0 {
            count -= 1
            headB = headB.Next
        }
    }

    for {
        if headA == headB {
            return headA
        }
        headA = headA.Next
        headB = headB.Next
    }
```

```
    return nil
}
```

已知单链表中存在环，求进入环中第一个节点

```
func detectCycle(head *ListNode) *ListNode {
    if head == nil {
        return nil
    }

    origin := head

    faster := head
    for {
        if faster == nil || faster.Next == nil {
            return nil
        }
        faster = faster.Next.Next
        if head.Next == nil {
            return nil
        }
        head = head.Next
        if faster == head {
            break
        }
    }

    headA, headB := origin, faster

    a, b := headA, headB
    len_a, len_b := 1, 1
    for {
        if a.Next == headB {
            break
        }
        len_a += 1
        a = a.Next
    }
    for {
        if b.Next == headB {
            break
        }
        len_b += 1
        b = b.Next
    }


    if len_a > len_b {
        count := len_a - len_b
        for count > 0 {
            count -= 1
            headA = headA.Next
        }
    } else if len_a < len_b {
         count := len_b - len_a
```

```
        for count > 0 {
            count -= 1
            headB = headB.Next
        }
    }

    for {
        if headA == headB {
            return headA
        }
        headA = headA.Next
        headB = headB.Next
    }

    return nil
}
```

[删除链表的节点]

```
func deleteNode(head *ListNode, val int) *ListNode {
    if head == nil {
        return head
    }
    if head.Val == val {
        return head.Next
    }
    if head.Next != nil && head.Next.Val == val {
        head.Next = head.Next.Next
    }

    deleteNode(head.Next, val)

    return head
}
```

3.2 堆

3.3 二叉树

二叉树结构

```
// Definition for a binary tree node.
type TreeNode struct {
    Val int
    Left *TreeNode
    Right *TreeNode
 }
```

[二叉搜索树的最近公共祖先]

```
struct TreeNode* lowestCommonAncestor(struct TreeNode* root, struct TreeNode* p,
struct TreeNode* q) {
    if (root == NULL || p == root || q == root) return root;
```

```
    if ((p->val < root->val && q->val > root->val) || (p->val > root->val && q->val < root->val))
        return root;

    if (p->val < root->val && q->val < root->val)
        return lowestCommonAncestor(root->left, p, q);

    if (p->val > root->val && q->val > root->val)
        return lowestCommonAncestor(root->right, p, q);

    return root;
}
```

二叉树的直径

二叉树的最大距离（即相距最远的两个叶子节点）

[二叉树的深度]

```
func maxDepth(root *TreeNode) int {
    result := 0
    if root == nil {
        return result
    }

    stack := []*TreeNode{}
    stack = append(stack, root)

    for {
        if len(stack) == 0 {
            break
        }
        line := []int{}
        now_stack := []*TreeNode{}

        for i := 0;  i < len(stack); i++ {
                line = append(line, stack[i].Val)
                if stack[i].Left != nil {
                    now_stack = append(now_stack, stack[i].Left)
                }
                if stack[i].Right != nil {
                    now_stack = append(now_stack, stack[i].Right)
                }
            }
        stack = now_stack
        result = result + 1
    }

    return result
}
```

二叉树的宽度

3.4 并查集

示例题目：[省份数量]

```go
var fa[210]int

func findXFa(x int) int {
    if x == fa[x] {
        return x
    } else {
        return findXFa(fa[x])
    }
}

func union(x int, y int) {
    fx := findXFa(x)
    fy := findXFa(y)

    if (fx == fy){
        return
    } else if fy > fx {
        fa[fy] = fx
    } else {
        fa[fx] = fy
    }
}

func findCircleNum(isConnected [][]int) int {
    n := len(isConnected)
    for i := 0; i < n; i++ {
        fa[i] = i
    }

    for i := 0; i < n; i++ {
        for j := 0; j < n; j++ {
            if i != j && isConnected[i][j] == 1 {
                union(i, j)
            }
        }
    }

    var result int
    for i := 0; i < n; i++ {
        if fa[i] == i {
            result += 1
        }
    }
    return result
}
```

3.5 树状数组

3.6 线段树

3.7 Trie树

3.8 二叉排序树

第四章 动态规划

4.1 01背包问题

```
package main
import "fmt"

type Bag struct {
    volume int
    value int
}

func max(a, b int) int {
    if a > b {
        return a
    }
    return b
}

func max_value(bags []*Bag ,capacity int) int {
    n := len(bags)
    var dp [100][100]int

    for i:= 1 ; i<n; i++ {
        for j:=0; j<=capacity; j++ {
            if j >= bags[i].volume {
                dp[i][j] = max(dp[i-1][j],dp[i-1][j-bags[i].volume] +
bags[i].value)
            } else {
                dp[i][j] = dp[i-1][j]
            }
        }
    }

    return dp[n-1][capacity]
}

func main () {
    bag1 := &Bag{1, 2}
    bag2 := &Bag{3, 4}
    bag3 := &Bag{2, 3}

    bags := []*Bag{bag1, bag2, bag3}
    capacity := 5
    fmt.Println(max_value(bags, capacity))
}
```

4.2 最长递增子序列

```
package main
import "fmt"

func max(a, b int) int {
    if a > b {
        return a
    }
    return b
}
```

```go
func lis(a[]int) int {
    var dp [101]int
    n := len(a)

    for i:=0; i<n; i++ {
        dp[i] = 1
        for j:=0; j<i; j++ {
            if a[i] > a[j] {
                dp[i] = max(dp[i], dp[j]+1)
            }
        }
    }
    return dp[n-1]
}

func main () {
    a := []int{1, 3, 2, 5, 3, 7, 4, 5}
    fmt.Println(lis(a))
}
```

4.3 最长公共子序列

```go
package main
import "fmt"

func max(a, b int) int {
    if a > b {
        return a
    }
    return b
}

func lcs(a, b []int) int {
    var dp [101][101]int
    n, m := len(a), len(b)

    for i:=1; i<=len(a); i++ {
        for j:=1; j<=len(b); j++ {
            if a[i-1] == b[j-1] {
                dp[i][j] = max(max(dp[i-1][j-1] + 1, dp[i-1][j]), dp[i][j-1])
            } else {
                dp[i][j] = max(dp[i-1][j], dp[i][j-1])
            }
        }
    }
    return dp[n][m]
}

func main () {
    a := []int{1, 2, 3, 4, 5}
    b := []int{3, 4, 4, 4, 5}
    fmt.Println(lcs(a, b))
}
```

4.4 最大字段和

```go
func maxSubArray(nums []int) int {
    sum, res := nums[0], nums[0]

    for i := 1; i < len(nums); i++ {
        if sum > 0 {
            sum += nums[i]
        } else {
            if nums[i] > sum {
                sum = nums[i]
            }
        }
        if sum > res {
            res = sum
        }
    }
    return res
}
```

4.5 树塔问题

```go
func minimumTotal(triangle [][]int) int {
    n := len(triangle)
    var dp [210][210]int

    for i:=n-1; i>=0; i-- {
        for j:=0; j<=i; j++ {
            if triangle[i][j] + dp[i+1][j] < triangle[i][j] + dp[i+1][j+1] {
                dp[i][j] = triangle[i][j] + dp[i+1][j]
            } else {
                dp[i][j] = triangle[i][j] + dp[i+1][j+1]
            }
        }
    }
    return dp[0][0]
}
```

4.6 两个字符串的编辑距离

第五章 搜索

5.1 二分查找

```go
package main
import "fmt"

func binarySearchIndex(arr []int, x int) int {
    st, ed := 0, len(arr)-1
    for st <= ed {
        mid := (st + ed)/2
        if arr[mid] == x {
            return mid
        }
        if arr[mid] > x {
            ed = mid-1
```

```go
        } else {
            st = mid+1
        }
    }
    return -1
}

func main () {
    var arr = []int{1,2,3,4,6,7,8,9,10}
    x := 5
    fmt.Println(binarySearchIndex(arr, x))
}
```

5.2 BFS

5.3 DFS

5.4 A*

5.5 三分

第六章 图算法

6.1 最短路径

6.2 二分图

6.3 拓扑排序

示例题目： [课程表 II]

```go
func findOrder(numCourses int, prerequisites [][]int) []int {
    in := make(map[int]int) // 点的入边数量
    edge := make(map[int][]int) // 点的下一个连接点

    for i := 0; i < len(prerequisites); i++ {
        prerequisite := prerequisites[i]
        edge[prerequisite[1]] = append(edge[prerequisite[1]], prerequisite[0])
        in[prerequisite[0]] += 1
    }

    queue := []int{}
    for i := 0; i < numCourses; i++ {
        if in[i] == 0 {
            queue = append(queue, i)
        }
    }

    result := []int{}
    var top int
    for len(queue) > 0 {
        top = queue[0]
        queue = queue[1:]
```

```
        result = append(result, top)
        for _, val := range edge[top] {
            in[val] -= 1
            if in[val] == 0 {
                queue = append(queue, val)
            }
        }
    }
    if len(result) == numCourses {
        return result
    }
    return []int{}
}
```

6.4 最小生成树

第七章 字符串算法

7.1 KMP

7.2 BM

7.3 Manacher

7.4 AC自动机

第八章 海量数据处理

8.1 布隆过滤器

8.2 Bitmap

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8.4 MapReduce